12TH CHEMISTRY · CHAPTER 29 · FULL CHAPTER NOTES

Spectroscopy–1

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION: ELECTROMAGNETIC RADIATION

Q.1 [Short] What is electromagnetic radiation? How are wavelength, frequency and speed related?

Light and other forms of electromagnetic radiation travel in the form of waves, which consist of oscillating electric and magnetic fields. These waves are defined by their wavelength (λ) and frequency (ν), and they travel at the speed of light ($c = 3.0 \times 10^{8}$ m s^{-1}).

$$c = \lambda \nu$$

The electromagnetic spectrum comprises radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays. Gamma rays have the highest energy and radio waves the lowest. The visible region is about 400–750 nm.

Q.2 [Short] State the two points of Planck's quantum theory and give the energy equation.

1. Only discrete amounts of energy may be emitted or absorbed by atoms and molecules. Quantum energy is the smallest quantity of electromagnetic radiation that can be released or absorbed.
2. The frequency of a radiation and the energy it carries are directly proportional to each other.

$$E = h\nu = hc/\lambda$$

E = energy of the radiation, h = Planck's constant (6.626×10^{-34} J s), ν = frequency of radiation.

29.1 INDEX OF HYDROGEN DEFICIENCY (IHD)

Q.3 [Short] What is the index of hydrogen deficiency (degree of unsaturation)?

The index of hydrogen deficiency (IHD), or degree of unsaturation, is a measure of the degrees of unsaturation in a compound. It is the difference in the number of **pairs** of hydrogen atoms between the compound and the acyclic alkane with the same number of carbon atoms. A compound has one degree of unsaturation for every two hydrogen atoms that are missing.

Example: 1-hexene and cyclohexane (both C_6H_{12}) have two hydrogens less than the saturated alkane C_6H_{14}, so each has IHD = 1. A double bond and a ring each count as one IHD; a triple bond counts as two.

Q.4 [Short] How is the IHD of a hydrocarbon calculated? Relate it to structure.

For a hydrocarbon CnHm: IHD = (2n + 2 − m) ÷ 2

Formula	IHD	Structure unit involved
C_nH_{2n+2}	0	open chain alkane only
C_nH_{2n}	1	1 double bond **or** 1 ring
C_nH_{2n-2}	2	2 double bonds, or 2 rings, or 1 double bond plus 1 ring, or 1 triple bond

Q.5 [Short] A compound C_6H_{10} has how many degrees of unsaturation?

The saturated alkane with six carbons is C_6H_{14}. C_6H_{10} has four hydrogen atoms (two pairs) less, so IHD = 2.

Q.6 [Short] Why is the IHD important?

Knowing the IHD of a molecule gives clues about its composition and structure, and it is found easily just from the formula. If IHD is 0 the compound has no rings and no double bonds. If IHD is 1 it has either one ring or one double bond. If IHD is 2 it could have two rings, two double bonds, one of each, or one triple bond. IHD thus gives useful information without complex analytical techniques, and it helps to assess which analytical technique would be better for the compound.

Quick Check 29.1

Q.7 [Short] a) Calculate the IHD for each of the following. i) C_6H_6 ii) C_6H_{12} iii) C_4H_4

For a hydrocarbon CnHm: IHD = (2n + 2 − m) ÷ 2

i) C_6H_6: IHD = (14 − 6) ÷ 2 = 4

ii) C_6H_{12}: IHD = (14 − 12) ÷ 2 = 1

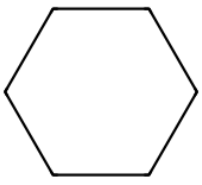
iii) C_4H_4: IHD = (10 − 4) ÷ 2 = 3

Q.8 [Short] b) Propose the structure of the above compounds.

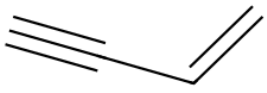
One possible structure for each: (i) benzene – one ring and three double bonds (IHD 4); (ii) cyclohexane – one ring (or hex-1-ene, one double bond); (iii) but-1-en-3-yne, CH_2=CH–C≡CH – one double and one triple bond (IHD 3).



(i) benzene



(ii) cyclohexane



(iii) but-1-en-3-yne

Q.9 [Short] **c) Hopanes are the waxy solids found in petroleum products as residues after refining the crude oil. A hopane has the formula $C_{30}H_{62}$, i) Find its IHD. ii) Predict whether this compound is saturated or not.**

$$\text{i) IHD} = (2 \times 30 + 2 - 62) \div 2 = 0$$

ii) IHD = 0, so the compound (as printed) is saturated: it has no rings and no multiple bonds.

29.2 ATOMIC SPECTRUM

Q.10 [Short] **What are atomic spectra and what do they prove?**

Atomic spectra are observed when atoms emit or absorb light of certain wavelengths. They provide evidence that electrons in atoms can only move between discrete energy levels.

Q.11 [Long] **Explain atomic emission and atomic absorption spectra.**

Emission spectrum: it is produced by heating a low-pressure gas. Heating excites electrons to higher energy levels. When an electron falls back to a lower level it emits a photon; each transition corresponds to a specific wavelength, seen as a spectral line. The emission spectrum is a set of **coloured lines on a black background**.

Absorption spectrum: it is produced by passing white light through a cool, low-pressure gas. Only photons with the exact energy needed to excite the ground-state electrons are absorbed. Each absorbed wavelength appears as a **dark line on a coloured background**. These lines are at the same positions as the lines in the emission spectrum of the same element, so an absorption spectrum is like a photographic negative of an emission spectrum.

Emission spectrum	Absorption spectrum
Excited electrons fall to lower levels and emit photons.	Ground-state electrons absorb photons and rise to higher levels.
Produced by heating a low-pressure gas.	Produced by passing white light through a cool, low-pressure gas.
Coloured lines on a black background.	Dark lines on a coloured background.

Q.12 [Short] **Why is an absorption spectrum called a photographic negative of an emission spectrum?**

Because the dark lines of the absorption spectrum appear at exactly the same wavelengths as the bright coloured lines of the emission spectrum of the same element: the element absorbs the same wavelengths that it emits.

Q.13 [Short] How do emission spectra help to deduce the electronic configuration of elements?

Each line in an emission spectrum is produced by an electron falling from a higher energy level to a lower one, so the lines show that electrons exist only in discrete energy levels (shells and sub-shells). From the series of lines, the energy levels occupied by the electrons, and therefore the electronic configuration of the element, can be deduced.

The chapter lists this outcome without a separate explanation; the answer follows from the text on discrete energy levels.

29.3 SPECTROSCOPY

Q.14 [Short] What is spectroscopy? What information does a spectrum give?

Spectroscopy deals with the production, measurement and interpretation of spectra arising from the interaction of electromagnetic radiation with matter. A spectrum gives information about atomic and molecular energy levels, molecular geometries, interactions of molecules and the nature of chemical bonds. It is used to measure the amount of a substance (quantitative analysis) or to identify an unknown sample (qualitative analysis).

Q.15 [Short] How can elements be identified by their atomic spectra? Give examples.

Each element produces a unique pattern of spectral lines, like a fingerprint; no two elements give the same set, so elements can be identified by their atomic spectra. Hydrogen gives a strong red line at 656 nm; sodium gives a yellow flame due to lines at 589 nm; mercury has most lines below 450 nm, giving blue light. Astronomers use atomic spectra to find the chemical composition of stars.

29.4 UV/VISIBLE SPECTROSCOPY

Q.16 [Short] What is UV-Vis spectroscopy? How is absorbance related to transmittance?

In UV-Vis spectroscopy, UV-visible light (200–800 nm) is passed through a sample and the transmittance is measured. The absorbance is found from $A = -\log(T)$. The spectrum shows the absorbance at various wavelengths, which depends on the electronic structure of the molecule. It is popular because it is versatile and easy to use.

Q.17 [Short] Which types of electrons are present in a compound? Compare the energies of their orbitals.

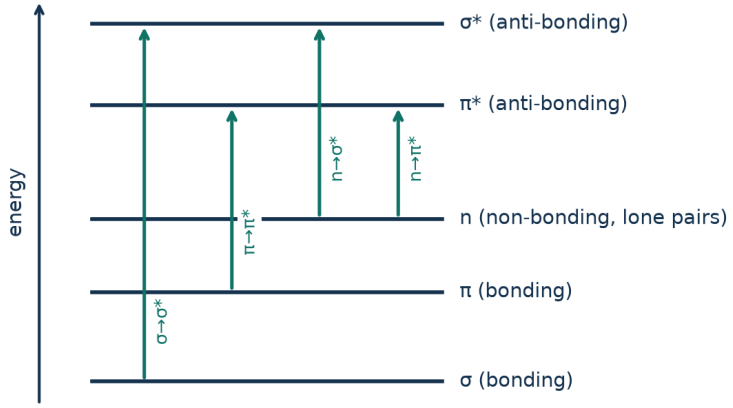
A compound may contain σ electrons (σ-bonds), π electrons (in unsaturated compounds only) and n electrons (non-bonded electrons). σ bonding orbitals are lower in energy than π bonding orbitals, which are lower than non-bonding orbitals. The anti-bonding orbitals σ* and π* are normally empty.

Q.18 [Long] Describe the electronic transitions possible in UV-Vis spectroscopy.

When radiation of the correct frequency is absorbed, an electron is excited from a filled orbital into an empty anti-bonding orbital. The important transitions are:

- π → π* (from pi bonding to pi anti-bonding orbitals)
- n → π* (from non-bonding to pi anti-bonding orbitals)
- n → σ* (from non-bonding to sigma anti-bonding orbitals)

σ → σ* needs the most energy and lies outside the 200–800 nm range. n → π* needs the least energy.



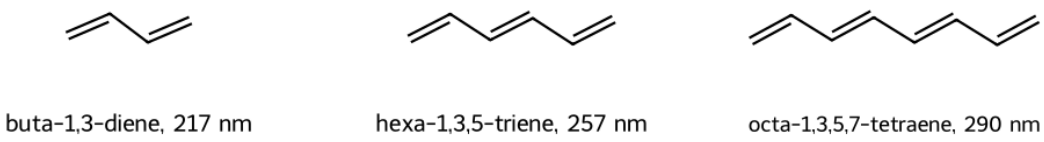
Electronic transitions from filled orbitals to anti-bonding orbitals

Q.19 [Short] What is λ_{max}? Explain with the spectrum of buta-1,3-diene.

Buta-1,3-diene, CH_2=CH–CH=CH_2, shows an absorption peak at 217 nm; the wavelength of maximum absorption is called λ_{max}. This is in the UV region, so the compound is colourless. It has no non-bonding electrons, so the only transition is π → π*.

Q.20 [Long] How does conjugation affect λ_{max} and the colour of a compound?

The value of λ_{max} depends on the extent of conjugation. The more conjugated a compound is, the more energy levels it has and the smaller the gaps between them. Less energy is needed to excite an electron, so light of longer wavelength is absorbed. Each extra conjugated double bond increases λ_{max} by about 30 to 40 nm. Highly conjugated compounds absorb in the visible region and are often brightly coloured.



buta-1,3-diene, 217 nm | hexa-1,3,5-triene, 257 nm | octa-1,3,5,7-tetraene, 290 nm

Q.21 [Short] How can you predict whether a molecule will absorb in the UV/visible region?

A molecule absorbs significantly in the UV-visible region (200–800 nm) if it has π electrons, especially conjugated double bonds, or non-bonding electrons next to a π bond (π → π* and n → π* transitions). Compounds with only σ bonds (alkanes) or only σ bonds and lone pairs (alcohols) do not absorb significantly in this range.

Q.22 [Short] How is the colour of a transition metal complex predicted from its UV/visible spectrum?

A transition metal complex absorbs certain wavelengths of visible light; the colour seen is the **complementary colour** of the light absorbed. So λ_{max} in the visible spectrum tells the colour absorbed, and the complex appears in the complementary colour. For example, a complex absorbing orange-red light appears blue, and one absorbing yellow-green light appears violet-purple.

The chapter lists this outcome but does not explain it; the answer uses the complementary-colour idea from Chapter 18 (Transition Metals).

Q.23 [Short] Why do green and red tomatoes have different colours?

A green tomato has more chlorophyll while a red ripe tomato contains lycopene. These compounds have different λ_{max} values, so each shows a specific colour in the visible range.

Quick Check 29.2

Q.24 [Short] a) Which type of electronic transitions require least energy in UV/Vis spectroscopy? Explain. i) σ→σ* ii) π→π* iii) n→σ* iv) n→π*

iv) n → π*. The non-bonding (n) orbital is the highest-energy filled orbital and π* is the lowest-energy empty orbital, so the gap between them is the smallest.

Q.25 [Short] b) Arrange the following in an increasing order of λ_{max}. i) a ketone whose C=O is conjugated with two C=C bonds (with an ethyl branch) ii) but-3-en-2-one, CH_2=CH–CO–CH_3 iii) hexa-3,5-dien-2-one, CH_2=CH–CH=CH–CO–CH_3

ii < iii < i. λ_{max} increases with the extent of conjugation. (ii) has only one C=C conjugated with C=O, so it absorbs at the shortest wavelength. (iii) and (i) both have two C=C bonds conjugated with C=O; (i) also carries an extra alkyl group on the conjugated chain, which shifts λ_{max} a little further.

The effect of the alkyl branch (placing i above iii) is beyond the textbook; from the chapter alone, i and iii have the same extent of conjugation and both absorb at a longer wavelength than ii.

Q.26 [Short] c) Will $CH_3CHOHCH_3$ show a significant absorption in the UV-Vis region.

No. Propan-2-ol has only σ bonds and lone pairs on oxygen; it has no π electrons. Its possible transitions (σ → σ* and n → σ*) need high energy and fall below 200 nm, outside the UV-Vis range (200–800 nm).

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29.5 INFRARED SPECTROSCOPY

Q.27 [Short] What is the basis of infrared spectroscopy? Name the modes of vibration.

All covalent bonds act like springs and vibrate in different ways (symmetric and asymmetric stretching, symmetric and asymmetric bending). The frequencies of these vibrations lie in the infra-red region. If a molecule is irradiated with infra-red energy that matches the natural vibration frequency of its bonds, it absorbs that energy and the amplitude of vibration increases. IR spectroscopy identifies compounds from these absorptions.

Q.28 [Short] Why are O_2 and H_2 IR inactive?

IR energy is absorbed only if a molecule has a permanent dipole that changes as it vibrates. Symmetrical, non-polar molecules such as O_2 or H_2 are therefore IR inactive.

Q.29 [Short] Which bonds absorb at higher wavenumbers and why?

Polar bonds such as O–H and N–H, bonds with smaller bond length and multiple bonds (e.g. HC≡CH) absorb at higher wavenumbers, because these bonds are stronger and require higher energy.

Q.30 [Short] What are the two regions of an IR spectrum?

An IR spectrum has two regions: the fingerprint region below 1500 cm^{-1} and the functional group region above 1500 cm^{-1}. The functional group region is enough to interpret the nature of an organic compound.

Q.31 [Long] Give the characteristic IR absorption ranges of common bonds.

Bond	Functional groups containing the bond	Absorption range / cm^{-1}
C–O	alcoholic, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3300 3200–3600

Q.32 [Long] Describe the IR absorptions of alcohols, amines, amides, carboxylic acids and ketones.

- **Alcohol:** strong O–H absorption at 3600–3200 cm^{-1}; C–H near 3000 cm^{-1}.

- **Amine:** N–H gives two weak absorptions at 3300 and 3400 cm^{-1}. The N–H signal is broad but not broader than the O–H signal of alcohols.
- **Amide:** N–H (two peaks) at 3200 and 3400 cm^{-1}; C=O at 1660 cm^{-1}.
- **Carboxylic acid:** strong C=O absorption at 1700 cm^{-1} with a broad, deep O–H absorption between 2500 and 3300 cm^{-1}, often distorted on its right side by C–H peaks.
- **Ketone:** strong, sharp C=O absorption around 1710 cm^{-1}.

Q.33 [Short] How can the IR spectra of propan-2-ol and propanone be told apart? (Example 29.1)

The spectrum with a strong, sharp absorption around 1710 cm^{-1} (C=O) and no O–H band is **propanone**. The spectrum with a strong absorption around 3200–3600 cm^{-1} (O–H) is **propan-2-ol**.

Q.34 [Short] A compound C_3H_6O shows a strong sharp peak at 1710 cm^{-1} and no band at 3200–3600 cm^{-1}. Deduce its structure.

$$\text{IHD} = (2 \times 3 + 2 - 6) \div 2 = 1$$

The peak at 1710 cm^{-1} is C=O (this accounts for the one degree of unsaturation) and there is no O–H. The compound is propanone (acetone), CH_3COCH_3. (Propanal, CH_3CH_2CHO, also fits these two facts.)

Q.35 [Short] How would the IR spectra of ethanol and phenol help to identify them?

Both show a strong, broad O–H absorption at 3200–3600 cm^{-1} and a C–O absorption at 1040–1300 cm^{-1}. Ethanol also shows alkane C–H absorption at 2850–2950 cm^{-1}, while phenol shows aromatic C=C absorptions at 1500–1680 cm^{-1}. Neither has a C=O peak.

Q.36 [Short] Give two everyday uses of infrared spectroscopy.

Breathalyser: infra-red radiation is passed through the exhaled breath; the characteristic bonds of ethanol are detected, and the higher the absorbance, the more ethanol is in the breath. **Vehicle emissions:** air sensors use IR to detect and measure pollutants such as carbon monoxide, carbon dioxide and unburnt hydrocarbons on motorways and in busy town centres.

Quick Check 29.2 (second box, infrared)

The textbook numbers two different boxes "Quick Check 29.2"; they are solved in the printed order.

Q.37 [Short] I. Which one of the infra-red spectra is that of butanone and which one is of butan-2-ol?

Spectrum A is butanone: it has a strong, sharp absorption near 1700 cm^{-1} (C=O) and no broad O–H band. **Spectrum B is butan-2-ol:** it has a strong, broad absorption at about 3200–3600 cm^{-1} (O–H) and no C=O peak.

Q.38 [Short] **ii. Compounds T and U are isomers with the molecular formula $C_3H_6O_2$. Suggest their structures based on the spectra shown below:**

T is propanoic acid, CH_3CH_2COOH: its spectrum shows a very broad O–H absorption (about 2500–3300 cm^{-1}) together with a strong C=O absorption near 1700 cm^{-1}. **U is an ester, methyl ethanoate CH_3COOCH_3** (ethyl methanoate, $HCOOCH_2CH_3$, is also possible): its spectrum has a strong C=O absorption (1710–1750 cm^{-1}) and C–O absorption but no O–H band.

29.6 MASS SPECTROMETRY

Q.39 [Short] **What is mass spectrometry?**

Mass spectrometry is an analytical technique used to identify unknown compounds. The molecules or atoms of a small sample are bombarded with high-energy electrons, which makes them lose an electron. The positive ion formed is detected and recorded as m/e (mass to charge ratio) against percentage abundance.

Q.40 [Short] **How does mass spectrometry give the relative abundance of isotopes?**

Mass spectrometry finds the relative abundance of isotopes experimentally. Relative abundance is the proportion of a particular isotope in the natural mixture of isotopes. The heights of the peaks show the proportion of each isotope. For example, Cl-35 and Cl-37 are 75% and 25%; boron-10 is 19.9% and boron-11 is 80.1%.

Q.41 [Long] **How is relative atomic mass calculated from isotopic data? Solve for oxygen and boron.**

$$\text{Relative atomic mass (Ar)} = \Sigma\ (\text{isotopic mass} \times \%\ \text{abundance}) \div 100$$

Oxygen (Example 29.2): ^{16}O 99.76%, ^{17}O 0.04%, ^{18}O 0.20%.

$$\text{Ar of oxygen} = [(99.76 \times 16) + (0.04 \times 17) + (0.20 \times 18)] \div 100$$
$$= 16.0044 = 16.00 \text{ (to 2 decimal places)}$$

Boron (Example 29.3): ^{10}B 19.9%, ^{11}B 80.1%.

$$\text{Ar of boron} = [(19.9 \times 10) + (80.1 \times 11)] \div 100$$
$$= 10.801 = 10.80 \text{ (to 2 decimal places)}$$

Q.42 [Short] **Chlorine is 75% Cl-35 and 25% Cl-37. Calculate its relative atomic mass.**

$$\text{Ar} = [(75 \times 35) + (25 \times 37)] \div 100 = 35.5$$

Q.43 [Short] What is the molecular ion peak? What information does it give?

When molecules are bombarded with highly accelerated electrons they lose an electron and form a positively charged **molecular ion** (M^+) with one unpaired electron. The peak mostly with the highest m/e value is the molecular ion peak; it gives the molecular mass of the compound. The molecular ion can fragment further into new ions, molecules and free radicals.

$$M + e^- \rightarrow M^+ + 2e^-$$

Q.44 [Short] A mass spectrum has its molecular ion peak at m/e 58. Is it propanal or butanal? (Example 29.4)

The molecular ion peak is at m/e = 58, which is the mass of the $CH_3CH_2CHO^+$ ion, so the spectrum is of **propanal**. Butanal, $CH_3CH_2CH_2CHO$, would give a molecular ion at m/e = 72.

Q.45 [Long] What is fragmentation? Explain with the fragments of hexane.

Different compounds may have the same molecular mass, so fragmentation is used to determine structure. Fragments appear by the formation of characteristic smaller ions or the loss of small molecules, e.g. peaks at 18 (H_2O), 29 ($C_2H_5^+$) and 44 (CO_2). Simple alkanes fragment by breaking C–C bonds.

Fragment	m/e	Fragment	m/e
CH_3^+	15	$C_4H_9^+$	57
$C_2H_5^+$	29	$C_5H_{11}^+$	71
$C_3H_7^+$	43	$C_6H_{13}^+$	85

Straight-chain hexane gives peaks at 86 (molecular ion, $C_6H_{14}^+$), 71 ($C_5H_{11}^+$), 57 ($C_4H_9^+$), 43 ($C_3H_7^+$), 29 ($C_2H_5^+$) and 15 (CH_3^+), due to breaking of the chain at different locations.

Q.46 [Long] What is the [M+1] peak? How does it give the number of carbon atoms?

The [M+1] peak is a small peak due to the natural abundance of carbon-13 (about 1.1% of all carbon atoms). The more carbon atoms in the molecule, the larger the [M+1] peak; e.g. it is higher for hexane than for ethane. The number of carbon atoms, n, is:

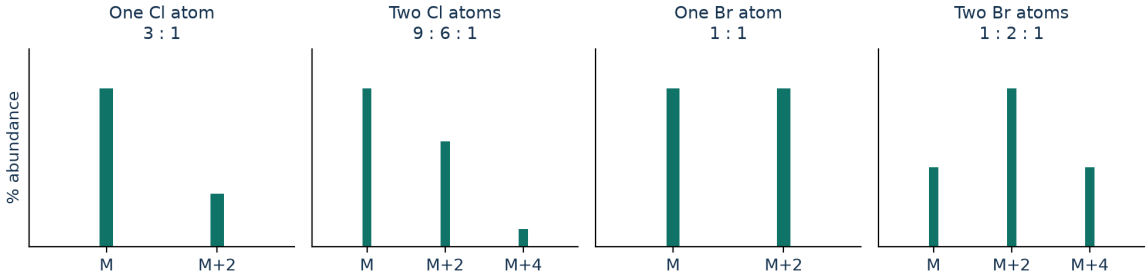
$$n = (100 \times \text{abundance of [M+1]}) \div (1.1 \times \text{abundance of M})$$

Example: M peak at m/e 58 with abundance 85 and [M+1] at m/e 59 with abundance 3 gives n = (100 × 3) ÷ (1.1 × 85) = 3.21, i.e. 3 carbon atoms.

Q.47 [Long] How do [M+2] and [M+4] peaks show the presence of chlorine and bromine?

Chlorine has two isotopes, ^{35}Cl and ^{37}Cl (3 : 1); bromine has ^{79}Br and ^{81}Br (1 : 1). Compounds containing them show [M+2] (and possibly [M+4]) peaks:

Atoms present	Peaks	Ratio of heights
One Cl	M (^{35}Cl), M+2 (^{37}Cl)	3 : 1
Two Cl	M (35+35), M+2 (35+37), M+4 (37+37)	9 : 6 : 1
One Br	M (^{79}Br), M+2 (^{81}Br)	1 : 1
Two Br	M (79+79), M+2 (79+81), M+4 (81+81)	1 : 2 : 1



Quick Check 29.3

Q.48 [Short] **a) Show all the possible fragments of i) C_4H_{10} ii) C_5H_{12} iii) C_7H_{16}**

Taking straight-chain alkanes, the C–C bonds break to give alkyl ions:

Compound	Molecular ion (m/e)	Fragment ions (m/e)
i) C_4H_{10}	$C_4H_{10}^+$ (58)	$C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)
ii) C_5H_{12}	$C_5H_{12}^+$ (72)	$C_4H_9^+$ (57), $C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)
iii) C_7H_{16}	$C_7H_{16}^+$ (100)	$C_6H_{13}^+$ (85), $C_5H_{11}^+$ (71), $C_4H_9^+$ (57), $C_3H_7^+$ (43), $C_2H_5^+$ (29), CH_3^+ (15)

Q.49 [Short] **b) The mass spectrum of CH_3Br is given below. [Peaks at m/e 15, 94 and 96] i) Identify the fragment against each peak. ii) What is the justification of two peaks at 94 and 96?**

i) m/e 15: CH_3^+; m/e 94: $CH_3{}^{79}Br^+$ (molecular ion, M); m/e 96: $CH_3{}^{81}Br^+$ ([M+2] peak).

ii) Bromine exists as two isotopes, ^{79}Br and ^{81}Br, in almost equal abundance. So there are two molecular ion peaks, M and [M+2], two units apart and of similar height (1 : 1).